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Main Theorem. *Let (M,ω) be a closed symplectic manifold of real dimension $2n$ and let $D = D_1 \cup \cdots \cup D_s \subset M$ be a symplectic simple normal crossings divisor. Then there exist a path ω_t, $t \in [0,1]$, of cohomologous symplectic forms with $\omega_0 = \omega$, along which D remains a symplectic simple normal crossings divisor, and an ω_1-tame almost complex structure J such that every $p \in D$ has a neighborhood U with smooth complex coordinates $(z_1,\ldots,z_n)$ centered at p for which, for some $1 \le r \le n$, the functions $z_1,\ldots,z_r$ are J-holomorphic and*

$$D \cap U = \{z_1 \cdots z_r = 0\}.$$

The proof combines the regularization theorem of Tehrani–McLean–Zinger [TMZ] with a relative simultaneous version of Donaldson's theorem [Do], using Auroux's vector-bundle generalization [Au], and then argues by pseudo-holomorphic structures on normal bundles constructed inductively by pseudo-holomorphic divisors. We begin by discussing the latter two notions.

1. Pseudo-holomorphic line bundles and divisors

Let (X,J) be an almost complex manifold. A $(0,1)$-connection on a complex vector bundle $E \to X$ is a $\mathbb{C}$-linear map $\bar\partial_E : C^\infty(E) \to C^\infty(E \otimes \Lambda^{0,1}_X)$ satisfying $\bar\partial_E(fs) = s \otimes \bar\partial_J f + f\bar\partial_E s$. It is a *$J$-holomorphic structure* if E locally has frames $e_1,\ldots,e_m$ with $\bar\partial_E e_i = 0$. Such a bundle will also be called *pseudo-holomorphic*. A morphism $\varphi : E \to F$ of pseudo-holomorphic vector bundles is a smooth complex-linear bundle map satisfying $\bar\partial_F(\varphi s) = (\varphi \otimes \mathrm{id})\bar\partial_E s$ for every smooth section s of E. An isomorphism is an invertible morphism. Equivalently, pseudo-holomorphic bundles have J-holomorphic transition functions, and their morphisms are represented in pseudo-holomorphic frames by matrices of J-holomorphic functions. Direct sums, tensor products and duals have the induced structures.

A pseudo-holomorphic vector bundle E has a natural almost complex structure on its total space. Indeed, in a pseudo-holomorphic frame let $z_1,\ldots,z_m$ be the linear fiber coordinates and declare the $(1,0)$-forms to be generated by the pullbacks of $\Lambda^{1,0}_X$ and $dz_1,\ldots,dz_m$. A pseudo-holomorphic change of frame preserves this subbundle. The resulting almost complex structure is the unique one which induces the given complex structure on every fiber and for which the projection $E \to X$ and all pseudo-holomorphic sections are pseudo-holomorphic.

A smooth codimension-two submanifold $Z \subset X$ is a *J-holomorphic divisor* if it is locally the transverse zero locus of a J-holomorphic function. A union $Z_1 \cup \cdots \cup Z_s$ is a *J-holomorphic snc divisor* if the branches through any point have J-holomorphic defining functions $f_1,\ldots,f_r$ such that $(f_1,\ldots,f_r)$ is a submersion. The smooth constant-rank theorem completes these functions to complex-valued smooth coordinates, so this is the local condition in the Main Theorem.

The following elementary fact is responsible for the rigidity of smooth pseudo-holomorphic divisors.

Lemma 1.1. *Let $f,g : U \to \mathbb{C}$ be J-holomorphic functions having the same transverse, positively oriented zero locus Z. Then f/g extends uniquely to a nowhere-vanishing J-holomorphic function on U.*

Proof. The assertion is local near a point $p \in Z$. Since g is a submersion, we may take $z_1 = g$ as one of a system of smooth complex-valued coordinates centered at p, so $Z = \{z_1 = 0\}$ and $\bar\partial_J z_1 = 0$. Write $z_1 = x_1 + iy_1$. Integrating the radial derivative of $f(tx_1, ty_1, z')$ from $t = 0$ to $t = 1$, and using $f|_Z = 0$, gives $f = az_1 + b\bar z_1$ for smooth functions a, b. Hence

$$0 = \bar\partial_J f = z_1 \bar\partial_J a + b\bar\partial_J \bar z_1 + \bar z_1 \bar\partial_J b.$$

Restricting to Z gives $b|_Z = 0$. Thus $b = O(|z_1|)$ and $f/z_1 = a + b\bar z_1/z_1$ extends continuously across Z, with value $a|_Z$. This value is nonzero by transversality and the orientation assumption. The quotient is locally bounded and J-holomorphic on $U \setminus Z$. Since Z has real codimension two, [HP, Thm. 4.1(b)], applied to $\bar\partial_J$, shows that it is a weak solution across Z; elliptic regularity then makes it smooth and J-holomorphic. □

Proposition 1.2. *For every smooth J-holomorphic divisor $Z \subset X$ there is a pseudo-holomorphic line bundle $\mathscr{L}_X(Z)$ with a pseudo-holomorphic section s_Z having transverse zero locus Z. The pair $(\mathscr{L}_X(Z), s_Z)$ is unique up to a unique pseudo-holomorphic isomorphism preserving the section. Conversely, every transverse pseudo-holomorphic section of a pseudo-holomorphic line bundle defines a smooth J-holomorphic divisor.*

Proof. Choose local J-holomorphic defining functions f_α for Z, taking $f_\alpha = 1$ on open sets disjoint from Z. By Lemma 1.1, the quotients f_α/f_β extend to nowhere-vanishing J-holomorphic functions. They are the transition functions of $\mathscr{L}_X(Z)$, and the f_α define its canonical section s_Z. If (L, s) is another such pair, the local quotients of s by s_Z extend by the same lemma and give the unique pseudo-holomorphic isomorphism taking s_Z to s. The converse follows by writing a section in a pseudo-holomorphic frame. □

For an integral combination $E = \sum_i a_i Z_i$ of smooth J-holomorphic divisors, set $\mathscr{L}_X(E) = \bigotimes_i \mathscr{L}_X(Z_i)^{\otimes a_i}$, with negative tensor powers interpreted as powers of the dual. In particular, for an snc divisor $Z = \bigcup_i Z_i$, the notation $\mathscr{L}_X(Z)$ means $\bigotimes_i \mathscr{L}_X(Z_i)$. Thus Proposition 1.2 and Lemma 1.1 are applied separately to the smooth branches; no assertion about arbitrary singular zero loci is being made.

The normal line bundle $N_{Z|X}$ of a smooth J-holomorphic divisor also has a natural pseudo-holomorphic structure. A local defining function f trivializes it by $df : N_{Z|X} \to Z \times \mathbb{C}$. If g is another defining function and $f = hg$, then h is nowhere-vanishing and J-holomorphic by Lemma 1.1, while $df|_{N_{Z|X}} = h\, dg|_{N_{Z|X}}$ along Z. These trivializations define a pseudo-holomorphic line bundle, denoted $\mathscr{N}_{Z|X}$. In the local frame e_f with $s_Z = fe_f$, the normal derivative of s_Z sends $v \in N_{Z|X}$ to $df(v)e_f$. This gives the canonical pseudo-holomorphic identification

$$\mathscr{L}_X(Z)|_Z \simeq \mathscr{N}_{Z|X}. \tag{1.1}$$

We also use restrictions, tensor products and duals of this isomorphism.

2. **A relative Donaldson theorem**

We need the following relative version of Auroux's vector-bundle generalization of Donaldson's theorem [Au, Thm. 2]. An ω-regularization is understood in the sense of [TMZ, Def. 2.12].

Theorem 2.1. *Let (X,ω) be a closed symplectic manifold with $[\omega]$ integral, and let $D \subset X$ be a symplectic snc divisor equipped with an ω-regularization. Let L_ω be a complex line bundle with $c_1(L_\omega) = [\omega]$, and let $L_1, \dots, L_q$ be complex line bundles on X. Then, for every sufficiently large integer N, there are smooth, transverse sections s_i of $L_i \otimes L_\omega^{\otimes N}$ such that, for $H_i = s_i^{-1}(0)$, the union $D \cup H_1 \cup \cdots \cup H_q$ is a symplectic snc divisor.*

Proof. For a set I of components of D, write D_I for their intersection and put $D_\emptyset = X$. We refer to the nonempty intersections D_I, which need not be connected, as the (closed) strata of D. Choose an ω-compatible almost complex structure J adapted to the regularization, in the sense that all D_I are J-invariant and the normal models split along them. Fix the metric $g = \omega(\,\cdot\,, J\,\cdot\,)$, a Hermitian connection ∇_L on L_ω with curvature $-2\pi i\omega$, and Hermitian metrics and connections on the L_i.

We recall the terminology and the part of Auroux's modified proof of [Do] used below. For a fixed Hermitian vector bundle E, a sequence s_k of sections of $E \otimes L_\omega^k$ is *asymptotically J-holomorphic* if

(2.1)
$$|s_k| = O(1), \quad |\nabla s_k| = O(k^{1/2}), \quad |\bar\partial_J s_k| = O(1), \quad |\nabla^2 s_k| = O(k), \quad |\nabla \bar\partial_J s_k| = O(k^{1/2}).$$

Here ∇ and the norm are induced by ∇_L and a fixed connection on E and Riemannian metric on X. The sequence is *η-transverse* on a J-invariant submanifold Y for some $\eta > 0$ if, whenever $x \in Y$ and $|s_k(x)| < \eta$, the restriction $\nabla s_k(x)|_{T_xY}$ has a right inverse of norm at most $\eta^{-1}k^{-1/2}$. Thus its zero locus in Y is transverse, with a quantitative lower bound on the normal derivative. We regard the zero section of the zero bundle as η-transverse.

Auroux's vector-bundle theorem [Au, Thm. 2] produces asymptotically holomorphic sequences of sections of $E \otimes L_\omega^k$ that are uniformly transverse, meaning η-transverse for some $\eta > 0$. The local transversality argument proceeds, in local trivializations of E, one summand at a time. If $E = E' \oplus \mathbb{C}$, let w_k and σ_k be asymptotically holomorphic sequences of sections of $E' \otimes L_\omega^k$ and L_ω^k, respectively, with w_k uniformly transverse. Perturbing σ_k by sums of localized Gaussian sections makes the restriction $\sigma_k|_{W_k}$ to $W_k = w_k^{-1}(0)$ uniformly transverse, and hence makes (w_k, σ_k) uniformly transverse. The perturbations may be taken arbitrarily small, uniformly in k, in the scaled norms

$$\|u\|_{C^1_k} = \sup_X \big(|u| + k^{-1/2}|\nabla u|\big). \tag{2.2}$$

This flexibility is what allows the componentwise construction to be repeated while retaining earlier transversality conditions.

We recall the local sections entering this construction. In Darboux coordinates $z = (z_1, \dots, z_n)$ centered at x, and in the radial trivialization of L_ω, the model section of L_ω^k is the Gaussian $\exp(-k|z|^2/4)$. Multiplying by a cutoff which equals one on the ball of radius $\frac{1}{2}k^{-1/3}$ and vanishes outside the ball of radius $k^{-1/3}$ gives a global peak section $\sigma_{k,x}$. On the cutoff region the Gaussian is $O(e^{-ck^{1/3}})$ for some $c > 0$, so the cutoff error is negligible to every fixed order. The resulting peak sections satisfy (2.1). Their centers are chosen in a $k^{-1/2}$-dense set $S = S_1 \cup \cdots \cup S_\ell$, where ℓ is independent of k and the points in each S_j are uniformly separated at scale $k^{-1/2}$. Following [Do], Auroux perturbs successively over the S_j. Gaussian decay controls interference within each step, while decreasing perturbation sizes preserve the transversality obtained in earlier steps.

We need two modifications of this construction, both already implicit in the proof of the vector-bundle theorem. First, suppose that the new component is to take values in $L_i \otimes L_\omega^k$ rather than L_ω^k. Near each center choose a unitary frame $e_{i,x}$ of L_i and replace $\sigma_{k,x}$ by $e_{i,x} \otimes \sigma_{k,x}$. Since L_i is fixed and X is compact, these frames can be chosen so that the connection matrix of L_i and its first derivative are uniformly bounded. The additional terms obtained by differentiating $e_{i,x}$ are therefore of lower order than the corresponding derivatives of $\sigma_{k,x}$. Hence all estimates in (2.1), as well as the Gaussian summation estimates, remain unchanged. This is the same local argument used by Auroux for a general fixed vector bundle.

Second, the construction can be made relative to a J-invariant symplectic submanifold Y. Suppose that w_k is uniformly transverse on both X and Y. First run the componentwise construction on Y, using centers in Y. Choose approximately holomorphic Darboux coordinates on Y at each center and extend them to Darboux coordinates on X adapted to Y. In these coordinates, Y is a complex coordinate subspace, and the ambient Gaussian peak section restricts to the Gaussian peak section used on Y. Thus the perturbation constructed on Y is the restriction of an ambient asymptotically holomorphic perturbation. It makes the new component transverse on $Y \cap w_k^{-1}(0)$, and hence makes the enlarged vector-bundle section transverse on Y.

One then runs the ambient componentwise construction. Its perturbation may be chosen so small in the norm (2.2) that the transversality already obtained on Y is preserved. Indeed, if a section is η-transverse, a sufficiently small C_k^1-perturbation is $\eta/2$-transverse. The same argument can be repeated for finitely many J-invariant submanifolds, taking each new perturbation small enough to preserve all conditions obtained before it. We shall apply this observation to the finite collection of strata of D.

For $A \subseteq \{1, \dots, q\}$ set $E_A = \bigoplus_{a \in A} L_a$ and, given sections $s_{a,k}$ of $L_a \otimes L_\omega^k$, write $s_{A,k} = \bigoplus_{a \in A} s_{a,k}$. We now construct the sections successively so that

$$s_{A,k}|_{D_I} \text{ is uniformly transverse for every } D_I \text{ and every nonempty } A. \tag{2.3}$$

Suppose $s_{1,k}, \dots, s_{m-1,k}$ have been constructed so that $s_{A,k}|_{D_I}$ is uniformly transverse for every stratum D_I and every nonempty $A \subseteq \{1, \dots, m-1\}$. The base case $m = 1$ gives $A = \emptyset$. Begin the construction of the m-th component with the zero section of $L_m \otimes L_\omega^k$. For every stratum D_I and every $A \subseteq \{1, \dots, m-1\}$, apply the relative componentwise construction to make $s_{A \cup \{m\},k}|_{D_I}$ uniformly transverse. When $A = \emptyset$, this means making $s_{m,k}|_{D_I}$ transverse.

After each such application choose the perturbation small enough to preserve all previously established conditions. There are finitely many pairs (I, A), so the process terminates with a positive uniform transversality constant. The total perturbation is a finite sum of asymptotically holomorphic perturbations and therefore still satisfies (2.1). It establishes (2.3) for every subset of $\{1, \dots, m\}$; conditions involving only the first $m-1$ components have not changed. Induction on m gives sections $s_{1,k}, \dots, s_{q,k}$ satisfying (2.3).

We finally verify that the zero loci have the required symplectic and orientation properties. Fix a stratum D_I, a nonempty subset A, and a point

$$p \in Z_{I,A,k} := D_I \cap \bigcap_{a \in A} s_{a,k}^{-1}(0).$$

The rescaled derivative

$$B_k = k^{-1/2}\nabla(s_{A,k}|_{D_I})(p) : T_pD_I \longrightarrow (E_A \otimes L_\omega^k)_p$$

is surjective and has a right inverse of uniformly bounded norm. Its complex-antilinear part satisfies $|B_k^{0,1}| = O(k^{-1/2})$ by (2.1). Thus the complex-linear part $B_k^{1,0}$ is also surjective for sufficiently large k: a right inverse for B_k gives one for $B_k^{1,0}$ by the usual Neumann-series argument. The same estimate shows that $\ker B_k$ is $O(k^{-1/2})$-close in the Grassmannian to the complex subspace $\ker B_k^{1,0}$. Since the complex Grassmannian is contained in the open set of symplectic planes, $\ker B_k$ is symplectic for large k. Moreover,

$$T_pZ_{I,A,k} = \ker B_k,$$

so $Z_{I,A,k}$ is a symplectic submanifold of the expected codimension.

It remains to compare orientations. For $t \in [0,1]$ consider the surjective maps $B_{k,t} = B_k^{1,0} + tB_k^{0,1}$. For large k all their kernels remain close to $\ker B_k^{1,0}$ and hence remain symplectic. At $t = 0$, the exact sequence of complex vector spaces

$$0 \longrightarrow \ker B_{k,0} \longrightarrow T_pD_I \xrightarrow{B_{k,0}} (E_A \otimes L_\omega^k)_p \longrightarrow 0$$

induces the complex, hence symplectic, orientation on $\ker B_{k,0}$. Both the intersection orientation and the symplectic orientation are constant along the path $t \mapsto \ker B_{k,t}$, so they agree at $t = 1$. Since this holds for every stratum D_I and every subcollection A, all intersections are ω-symplectic and positively oriented. Thus $D \cup H_1 \cup \cdots \cup H_q$ is an ω-symplectic snc divisor. This proves the theorem.

For later use, note that the intersection tangent spaces are uniformly $O(k^{-1/2})$-close to J-complex subspaces. Hence, for any other fixed symplectic form taming J, the same divisors are symplectic snc for all sufficiently large k. □

3. **Regularizations and the local normal model**

We use the normal-bundle charts of a regularization and their compatibility under passage between strata. Let $D = D_1 \cup \cdots \cup D_r$ be a symplectic snc divisor in the symplectic $2n$-manifold (X, ω). For $I \subseteq \{1, \ldots, r\}$ write $D_I = \bigcap_{i\in I} D_i$, with $D_\emptyset = X$, and

$$N_I = N_{D_I|X} = \bigoplus_{i\in I} N_i|_{D_I}, \qquad N_i = N_{D_i|X}.$$

An ω-regularization Ψ in the sense of [TMZ] consists, in particular, of compatible diffeomorphisms $\Psi_I : \mathcal{N}_I \to U_I$ from neighborhoods of the zero sections in N_I to neighborhoods of D_I in X. The compatibility conditions identify the normal-bundle charts associated to nested strata. Moreover, [TMZ, Thm. 2.13] states that, after deforming ω through cohomologous symplectic forms for which D remains a symplectic snc divisor, such a regularization exists.

We first recall the local symplectic form used in a regularization. Let (Y, ω_Y) be a symplectic manifold, let $\pi : N \to Y$ be a symplectic vector bundle with fiberwise form Ω, and choose a connection on N. Writing Ω_∇ for the 2-form extension of Ω which vanishes on the horizontal distribution and ζ for the fiberwise radial vector field, set

$$\omega_N = \pi^*\omega_Y + \frac{1}{2}d\big(\iota_\zeta\Omega_\nabla\big). \tag{3.1}$$

This is symplectic near the zero section and, with respect to the canonical splitting $T_N|_Y = TY \oplus N$, satisfies

$$\omega_N|_{T_N|_Y} = \omega_Y \oplus \Omega. \tag{3.2}$$

The regularization maps are symplectomorphisms from forms of this type to the restriction of ω to U_I.

Now let J_Y be an ω_Y-tame almost complex structure on Y, and suppose that the summands of N have pseudo-holomorphic line-bundle structures over (Y, J_Y) whose fiberwise complex structures are tamed by Ω. Their direct sum is a pseudo-holomorphic vector bundle, and the construction in Section 1 gives an almost complex structure J_N on its total space. Along the zero section,

$$J_N|_{T_N|_Y} = J_Y \oplus J_{\mathrm{fib}}.$$

Together with (3.2), this shows that J_N is tamed by ω_N along the zero section, and hence on a neighborhood of it by openness of tameness.

Applying this construction to N_I and pushing forward by Ψ_I produces an almost complex structure near D_I. In a local pseudo-holomorphic frame of $N_I = \bigoplus_{i\in I} N_i|_{D_I}$, let z_i be the linear coordinate on the i-th summand. The functions z_i are J_{N_I}-holomorphic, and

$$\Psi_I^{-1}(D_i) = \{z_i = 0\}, \qquad i \in I.$$

Thus, provided the normal-bundle structures agree on overlaps of the regularization charts, the pushed-forward structure makes D a J-holomorphic snc divisor.

We record the compatibility needed below. For $0 \le k \le n-1$, let $U(k)$ be the regularized neighborhood of the union of strata of complex dimension at most k, and set

$$D(k) = D \cap U(k), \qquad D_i(k) = D_i \cap U(k).$$

An almost complex structure on $D(k)$ means a compatible collection of almost complex structures on the smooth branches $D_i(k)$ and all their intersections.

An almost complex structure J on $U(k)$ is *Ψ-regularized* if, for every regularization chart $\Psi_I : \mathcal{N}_I \to U_I$ contained in $U(k)$, the pullback $\Psi_I^* J$ is the total-space almost complex structure defined by $J|_{D_I}$ and pseudo-holomorphic structures on the summands of $N_I = \bigoplus_{i\in I} N_i|_{D_I}$.

Lemma 3.1. *Fix a regularization Ψ, and set $U(-1) = \emptyset$. Suppose that a Ψ-regularized almost complex structure has already been constructed on $U(k-1)$. Let $\bar{J}$ be a compatible almost complex structure on $D(k)$ whose germ along $D(k-1)$ is induced by this structure. Suppose moreover that, for every i, the complex normal line bundle $N_i|_{D_i(k)}$ has a $\bar{J}$-holomorphic structure extending the structure on $N_i|_{D_i(k-1)}$ induced by the model on $U(k-1)$.*

Then, after shrinking the regularization domains, these data determine an almost complex structure J on $U(k)$ such that $\Psi_I^* J$ is the total-space almost complex structure defined by $\bar{J}|_{D_I}$ and the pseudo-holomorphic structures on the summands of N_I. Conversely, every almost complex structure of this form induces the stated compatible data on $D(k)$ and the normal line bundles.

Proof. Over a stratum D_I, take the direct sum of the prescribed pseudo-holomorphic structures on $N_i|_{D_I}$, $i \in I$, form the associated total-space almost complex structure on N_I, and push it forward by Ψ_I. If $D_{I'} \subset D_I$, the compatibility maps in the regularization identify

the restriction of this normal model with the model constructed from $N_{I'}$. On the part lying over $D(k-1)$ this is the compatibility assumed in the statement. Hence the pushed-forward structures agree on the overlaps of the charts Ψ_I, after their domains are shrunk, and define J on $U(k)$.

Conversely, pulling such a J back by Ψ_I and restricting to the zero section gives $\bar{J}|_{D_I}$. Restriction to each normal summand gives its pseudo-holomorphic line-bundle structure. Compatibility of these structures under inclusions of strata follows from the compatibility of the regularization maps. □

An almost complex structure obtained in this way will be called Ψ-*regularized*. The preceding discussion shows that a Ψ-regularized structure makes $D \cap U(k)$ a J-holomorphic snc divisor. It also shows that if $\bar{J}$ is tamed by the restrictions of ω to all strata and the fiberwise complex structures are tamed by the normal symplectic forms, then J is ω-tame after shrinking $U(k)$.

We finally recall the standard extension used at the end of the proof.

Lemma 3.2. *Let Ψ be an ω-regularization of D, and let J_U be a Ψ-regularized almost complex structure on a neighborhood U of D. Assume that its restriction to every stratum is tamed by the corresponding restriction of ω, and that the fiberwise complex structures on the normal line bundles are tamed by their normal symplectic forms. After shrinking U, the structure J_U extends to an ω-tame almost complex structure on X.*

Proof. By (3.2) and openness of tameness, J_U is ω-tame on a smaller neighborhood of D. The bundle of ω-tame linear complex structures has nonempty contractible fibers. Using the convex parametrization of these fibers from [MS, Prop. 2.5.13, Proof 3] and a cutoff function, one extends J_U from a still smaller neighborhood of D to an ω-tame almost complex structure on all of X. □

4. **Proof of the Main Theorem**

Proof. The construction has two distinct parts. We first enlarge D so that every normal line on every stratum has a divisor representative avoiding that stratum. We then use these representatives to construct compatible pseudo-holomorphic normal-bundle structures by induction on the dimension of the strata.

Write $X = M$, $\dim_{\mathbb{R}} X = 2n$, and enumerate the components of D as $D_1, \dots, D_r$.

Step I. *Auxiliary divisors.* For each $i = 1, \dots, r$, choose a complex line bundle L_i with a smooth section s_{i0} whose transverse, positively oriented zero locus is D_i. The standard tubular-neighborhood construction gives such a pair, with

$$c_1(L_i) = \mathrm{PD}[D_i].$$

Put $L_0 = \mathbb{C}_X$ and, for notational purposes, $D_0 = \emptyset$ and $s_{00} = 1$.

Possibly replacing ω by a cohomologous symplectic form along a path for which D remains symplectic snc, we may assume by [TMZ, Thm. 2.13] that D admits an ω-regularization.

Choose a rational symplectic form $\omega_{\mathbb{Q}}$ sufficiently close to ω. After a sufficiently small regularizing deformation, still denoted $\omega_{\mathbb{Q}}$, the divisor D admits an $\omega_{\mathbb{Q}}$-regularization. We may choose an $\omega_{\mathbb{Q}}$-compatible almost complex structure J_0 adapted to this regularization

and tamed by ω. Choose $a \in \mathbb{N}$ such that $a[\omega_{\mathbb{Q}}]$ is integral, and let P be a line bundle with $c_1(P) = a[\omega_{\mathbb{Q}}]$.

Apply Theorem 2.1, using J_0 as the auxiliary almost complex structure, to $n+1$ copies of each of the $r+1$ line bundles

$$L_0, L_1, \dots, L_r$$

on the integral symplectic manifold $(X, a\omega_{\mathbb{Q}})$. For sufficiently large N, it gives, for $i = 0, \dots, r$ and $\mu = 1, \dots, n+1$, sections $s_{i\mu}$ of

$$L_{i\mu} := L_i \otimes P^{\otimes N}$$

whose zero loci $H_{i\mu}$, together with D, form an $a\omega_{\mathbb{Q}}$-symplectic snc divisor. By the final observation in the proof of Theorem 2.1, after increasing N this divisor is also ω-symplectic snc. Thus

$$\widetilde{D} = D \cup \bigcup_{\substack{0 \le i \le r \\ 1 \le \mu \le n+1}} H_{i\mu} \tag{4.1}$$

is an ω-symplectic snc divisor. We refer to the nonempty intersections $\widetilde{D}_A$, which may be disconnected, as strata. All dimensions of strata below are complex.

Set $Q = P^{\otimes N}$ and write $\delta_i = [D_i]$, with $\delta_0 = 0$, and $h = \mathrm{PD}(c_1(Q))$. For $\mu > 0$, we regard all $L_{i\mu}$ as the same line bundle $L_i \otimes Q$; the index μ distinguishes the sections $s_{i\mu}$ and their zero loci. Thus

$$[H_{i\mu}] = \delta_i + h, \qquad L_{i\mu} = L_i \otimes Q.$$

We include the original components in this notation by setting

$$H_{i0} = D_i, \qquad L_{i0} = L_i, \qquad 1 \le i \le r.$$

For $\mu, \nu > 0$ we then have

$$L_{i\mu} = L_{i\nu} \quad (0 \le i \le r), \qquad L_{i\mu} \otimes L_{0\nu}^{\vee} \simeq L_{i0} \quad (1 \le i \le r), \tag{4.2}$$

where the second identification is induced by $Q \otimes Q^{\vee} \simeq \mathbb{C}_X$. Let

$$\Omega = \{(i,0) \mid 1 \le i \le r\} \cup \{(i,\mu) \mid 0 \le i \le r,\ 1 \le \mu \le n+1\}$$

index the hypersurfaces occurring in (4.1). For $\alpha = (i,\mu) \in \Omega$, write $\widetilde{D}_\alpha = H_{i\mu}$, $L_\alpha = L_{i\mu}$ and $s_\alpha = s_{i\mu}$. For $A \subseteq \Omega$, put

$$\widetilde{D}_A = \bigcap_{\alpha \in A} \widetilde{D}_\alpha, \qquad \partial \widetilde{D}_A = \bigcup_{\beta \notin A} \widetilde{D}_{A \cup \{\beta\}}.$$

Let $N_\alpha = N_{\widetilde{D}_\alpha | X}$. Transversality of s_α gives an orientation-preserving isomorphism

$$\nabla s_\alpha : N_\alpha \xrightarrow{\simeq} L_\alpha|_{\widetilde{D}_\alpha}, \tag{4.3}$$

independent of the chosen connection along the zero locus.

The $n+1$ copies ensure the existence of the successors used below. Since $\dim_{\mathbb{C}} X = n$, at most n distinct components of an snc divisor meet. Consequently, for every nonempty stratum $\widetilde{D}_A$ and every family $\{\widetilde{D}_{(i,1)}, \dots, \widetilde{D}_{(i,n+1)}\}$, at least one member has label outside A.

Step II. *Successor divisors.* We use the preceding redundancy to represent each normal line bundle $N_\alpha|_{\widetilde{D}_A}$ by the restriction of a divisor none of whose components contains $\widetilde{D}_A$. The representative changes upon restriction to a deeper stratum contained in its support.

Fix $A \subseteq \Omega$ with $\widetilde{D}_A \neq \emptyset$ and $\alpha = (i,\mu) \in A$. Endow $\{0,\dots,n+1\}$ with its cyclic order. For $\alpha = (i,\mu) \in A$, let $\nu_\alpha(A)$ and $\kappa_\alpha(A)$ be the first positive indices encountered after μ such that

$$(i,\nu_\alpha(A)) \notin A, \qquad (0,\kappa_\alpha(A)) \notin A,$$

respectively. These indices exist because $|A| \leq n$.

Define the formal *successor divisor*

$$\widetilde{D}^\dagger_{\alpha,A} = \begin{cases} \widetilde{D}_{(i,\nu_\alpha(A))} - \widetilde{D}_{(0,\kappa_\alpha(A))}, & \mu = 0, \\ \widetilde{D}_{(i,\nu_\alpha(A))}, & \mu > 0. \end{cases} \tag{4.4}$$

It is homologous to $\widetilde{D}_\alpha$, and every component occurring on the right-hand side has label outside A. Let $L_{\widetilde{D}^\dagger_{\alpha,A}}$ denote the line bundle obtained from $\widetilde{D}^\dagger_{\alpha,A}$ by replacing each $\widetilde{D}_\beta$ by L_β. The identifications in (4.2) give a canonical isomorphism

$$L_{\widetilde{D}^\dagger_{\alpha,A}} \simeq L_\alpha. \tag{4.5}$$

Combining (4.3) and (4.5) gives an isotopy class of orientation-preserving isomorphisms

$$N_\alpha|_{\widetilde{D}_A} \simeq L_{\widetilde{D}^\dagger_{\alpha,A}}|_{\widetilde{D}_A}. \tag{4.6}$$

In Step III the isotopy class of this isomorphism defines framings of the normal line bundle and the divisor line bundle associated to $\widetilde{D}_A|_{\widetilde{D}^\dagger_{\alpha,A}}$. We abbreviate $\widetilde{D}^\dagger_\alpha = \widetilde{D}^\dagger_{\alpha,\{\alpha\}}$.

Step III. *Regularization, framings and induction data.* Possibly replacing ω once more by a cohomologous symplectic form along a path for which $\widetilde{D}$ remains symplectic snc, we may assume by [TMZ, Thm. 2.13] that $\widetilde{D}$ admits an ω-regularization $\Psi = \{\Psi_A\}$.

Below we use the following topological extension statement. A *framing* of an oriented $\mathbb{R}^2$-bundle L relative to an oriented $\mathbb{R}^2$-bundle $L^\square$ over the same base is an isotopy class of orientation-preserving isomorphisms $\chi : L^\square \to L$. If framings of L and $\widehat{L}$ relative to the same $L^\square$ are represented by χ and $\widehat{\chi}$, an isomorphism $\varphi : L \to \widehat{L}$ is *framed* if $\widehat{\chi}^{-1}\varphi\chi$ is isotopic to the identity of $L^\square$. Now suppose a framed isomorphism $\varphi : L|_V \to \widehat{L}|_V$ is given on a neighborhood V of a closed subset $C \subset X$. After shrinking V around C, it extends to a framed isomorphism over X. Indeed, choose an isotopy a_t from the identity of $L^\square|_V$ to $\widehat{\chi}^{-1}\varphi\chi$ and a function $\rho : X \to [0,1]$ supported in V and equal to one near C. The automorphism $a_{\rho(x)}$ on V, extended by the identity outside V, gives the extension after conjugation by the framings.

For $0 \leq k \leq n-1$, let $U(k)$ be the regularized neighborhood of the union of strata of $\widetilde{D}$ of dimension at most k, and set $\widetilde{D}(k) = \widetilde{D} \cap U(k)$, $\widetilde{D}_\alpha(k) = \widetilde{D}_\alpha \cap U(k)$. Suppose temporarily that an almost complex structure $\bar{J}$ has been constructed on $\widetilde{D}(k)$ making the union of proper substrata of every $\widetilde{D}_A \subset \widetilde{D}(k)$ into a $\bar{J}|_{\widetilde{D}_A}$-holomorphic snc divisor. For $\alpha \in A$ define

$$\mathscr{L}^\dagger_{\alpha,A} = \mathscr{L}_{\widetilde{D}_A}\big(\widetilde{D}^\dagger_{\alpha,A}|_{\widetilde{D}_A}\big). \tag{4.7}$$

Here we use the notation for integral combinations of divisors introduced in Section 1.

We frame the normal line bundle and the line bundle in (4.7) by the same smooth bundle

$$L^\square_\alpha = L_\alpha|_{\widetilde{D}_\alpha}.$$

The inverse of (4.3) gives a framing $\chi^{\mathscr{N}}_{\alpha} : L^{\square}_{\alpha} \to N_{\alpha}$. On $\widetilde{D}_A$, the fixed isomorphism (4.5), together with the defining sections of its divisor factors, gives a framing $\chi^{\mathscr{L}}_{\alpha,A} : L^{\square}_{\alpha}|_{\widetilde{D}_A} \to \mathscr{L}^{\dagger}_{\alpha,A}$ of the underlying smooth bundle. Thus the smooth isomorphism (4.6) has a distinguished isotopy class.

The induction will construct pseudo-holomorphic structures $\mathscr{N}_{\alpha,k}$ on $N_{\alpha}|_{\widetilde{D}_{\alpha}(k)}$ and $L^{\square}_{\alpha}$-framed pseudo-holomorphic isomorphisms

$$\varphi_{\alpha,A} : \mathscr{N}_{\alpha,k}|_{\widetilde{D}_A} \xrightarrow{\simeq} \mathscr{L}^{\dagger}_{\alpha,A}, \qquad \alpha \in A, \quad \dim_{\mathbb{C}} \widetilde{D}_A \le k. \tag{4.8}$$

Their compatibility records the advancement of successors. Let $B = A \cup \{\beta\}$ and write $\alpha = (i, \mu)$. Put

$$\beta_+ = (i, \nu_{\alpha}(A))$$

and, if $\mu = 0$, also put

$$\beta_- = (0, \kappa_{\alpha}(A)).$$

Whenever a successor component is encountered, we use the canonical isomorphism

$$\mathscr{L}_{\widetilde{D}_A}(\widetilde{D}_{\beta})|_{\widetilde{D}_B} \simeq \mathscr{N}_{\beta,k}|_{\widetilde{D}_B}$$

from (1.1). The compatibility conditions are as follows.

(1) If $\beta \neq \beta_+$ and, when $\mu = 0$, also $\beta \neq \beta_-$, then $\mathscr{L}^{\dagger}_{\alpha,A}|_{\widetilde{D}_B} = \mathscr{L}^{\dagger}_{\alpha,B}$, and

$$\varphi_{\alpha,A}|_{\widetilde{D}_B} = \varphi_{\alpha,B}.$$

(2) If $\mu > 0$ and $\beta = \beta_+$, then

$$\varphi_{\alpha,A}|_{\widetilde{D}_B} = \varphi^{-1}_{\beta,B} \circ \varphi_{\alpha,B}.$$

(3) If $\mu = 0$ and $\beta = \beta_+$, then

$$\varphi_{\alpha,A}|_{\widetilde{D}_B} = \left(\varphi^{-1}_{\beta,B} \otimes \mathrm{id}\right) \circ \varphi_{\alpha,B}.$$

(4) If $\mu = 0$ and $\beta = \beta_-$, then

$$\varphi_{\alpha,A}|_{\widetilde{D}_B} = \left(\mathrm{id} \otimes \varphi^{\vee}_{\beta,B}\right) \circ \varphi_{\alpha,B}.$$

The framing isomorphisms define the compatible reference system

$$r_{\alpha,A} = \chi^{\mathscr{L}}_{\alpha,A} \circ (\chi^{\mathscr{N}}_{\alpha}|_{\widetilde{D}_A})^{-1}.$$

A compatible system is called *framed* if it is isotopic to this system through compatible systems of smooth bundle isomorphisms.

Thus the isomorphism simply restricts until a current successor is met. Its divisor line then restricts to its normal line bundle, which is replaced by the next successor using the already constructed isomorphism for that component. All identifications come from (4.2) and (1.1), and preserve the chosen framings.

These rules are compatible under iteration: the successor isomorphisms cancel telescopically, separately in the two tensor factors when $\mu = 0$. Thus the induced identification on a deeper stratum is independent of the chain of inclusions.

Step IV. *Induction over the strata.* We construct, for $k = 0, \dots, n-1$, a Ψ-regularized, ω-tame almost complex structure $J(k)$ on $U(k)$, and denote its restriction to $\widetilde{D}(k)$ by $\bar{J}(k)$. At the same time, we construct pseudo-holomorphic structures $\mathscr{N}_{\alpha,k}$ on the normal line bundles over $\widetilde{D}_{\alpha}(k)$ and a framed compatible system of isomorphisms (4.8). After shrinking

the regularization domains when necessary, all these data extend those constructed at the previous stage.

For $k = 0$, the strata are points, and the required normal-bundle structures and framed isomorphisms can be chosen pointwise with tame fiberwise complex structures. Lemma 3.1 then produces the corresponding almost complex structure $J(0)$ on $U(0)$. This starts the induction.

Assume the data have been constructed on $\widetilde{D}(k-1)$ and let $\widetilde{D}_A$ be a k-dimensional stratum. Its boundary $\partial\widetilde{D}_A$ lies in $\widetilde{D}(k-1)$. By the local-model Lemma 3.1, the already constructed almost complex structures and pseudo-holomorphic normal bundles determine an ω-tame regularized almost complex structure on a neighborhood of $\partial\widetilde{D}_A$ in $\widetilde{D}_A$. Extend its restriction to $T_{\widetilde{D}_A}$ over all of $\widetilde{D}_A$, using the contractibility of the space of $\omega|_{\widetilde{D}_A}$-tame almost complex structures. For distinct k-dimensional strata $\widetilde{D}_A$ and $\widetilde{D}_{A'}$, the loci $\widetilde{D}_A \setminus \partial\widetilde{D}_A$ and $\widetilde{D}_{A'} \setminus \partial\widetilde{D}_{A'}$ are disjoint. The extensions may therefore be chosen independently away from the prescribed boundary neighborhoods. They define $\bar{J}(k)$.

Fix now $\alpha \in A$. On each proper stratum $\widetilde{D}_B \subset \partial\widetilde{D}_A$, the compatibility rule of Step III expresses the required restriction of $\varphi_{\alpha,A}$ in terms of the isomorphisms already constructed on $\widetilde{D}(k-1)$. By the induction hypothesis, these prescriptions agree on intersections and define a single $L_\alpha^{\square}$-framed pseudo-holomorphic isomorphism

$$\varphi^{\partial}_{\alpha,A} : \mathscr{N}_{\alpha,k-1}|_{\partial\widetilde{D}_A} \xrightarrow{\simeq} \mathscr{L}^{\dagger}_{\alpha,A}|_{\partial\widetilde{D}_A}. \tag{4.9}$$

We apply the boundary prescriptions to one compatible framing isotopy from the preceding stage, using it for every A. This proves framedness over the whole boundary $\partial\widetilde{D}_A$. For every $B \supsetneq A$, the regularization supplies a retraction from a neighborhood of $\widetilde{D}_B$ in $\widetilde{D}_A$ to $\widetilde{D}_B$. These retractions are pseudo-holomorphic for the regularized models. By compatibility of the regularization and the induction data, the resulting pullbacks agree on overlaps and define an $L_\alpha^{\square}$-framed pseudo-holomorphic isomorphism on a neighborhood of $\partial\widetilde{D}_A$ in $\widetilde{D}_A$.

The two underlying smooth bundles over $\widetilde{D}_A$ have the framings defined in Step III. The framed extension statement therefore extends $\varphi^{\partial}_{\alpha,A}$, after shrinking its neighborhood, to a smooth $L_\alpha^{\square}$-framed isomorphism

$$\varphi_{\alpha,A} : N_\alpha|_{\widetilde{D}_A} \xrightarrow{\simeq} \mathscr{L}^{\dagger}_{\alpha,A}$$

over all of $\widetilde{D}_A$. Here we extend the boundary framing isotopy to a smaller neighborhood, with initial map the restriction of $r_{\alpha,A}$ and final map the isomorphism constructed above, and use this isotopy in the cutoff construction. Define the pseudo-holomorphic structure on $N_\alpha|_{\widetilde{D}_A}$ by pulling back that of $\mathscr{L}^{\dagger}_{\alpha,A}$ through $\varphi_{\alpha,A}$. Since $\varphi_{\alpha,A}$ is framed, it is orientation-preserving. The complex orientation of the resulting structure on the $\mathbb{R}^2$-bundle N_α therefore agrees with its symplectic orientation; in real dimension two this implies that the fiberwise complex structure is tamed by the normal symplectic form.

By construction the pseudo-holomorphic structure on $N_\alpha|_{\widetilde{D}_A}$ agrees with $\mathscr{N}_{\alpha,k-1}$ near the boundary, makes $\varphi_{\alpha,A}$ pseudo-holomorphic, and satisfies the compatibility rule on every proper stratum. Doing this for all $\alpha \in A$ and all k-dimensional strata completes the compatible data on $\widetilde{D}(k)$. By Lemma 3.1, these data determine a Ψ-regularized, ω-tame almost complex structure $J(k)$ on $U(k)$. Since they agree with the preceding data near $\widetilde{D}(k-1)$,

the regularization domains may be shrunk so that $J(k)$ extends $J(k-1)$. This proves the induction step.

At $k = n-1$ we have therefore obtained a Ψ-regularized, ω-tame almost complex structure $J(n-1)$ on $U(n-1)$. Its restriction $\bar{J}(n-1)$ to the whole divisor $\widetilde{D}$ is compatible with the pseudo-holomorphic structures constructed on all normal line bundles.

Step V. *Extension to X and local coordinates.* Set $U = U(n-1)$ and $J_U = J(n-1)$. By Lemma 3.2, after shrinking U the almost complex structure J_U extends to an ω-tame almost complex structure J on all of X, without changing it on a smaller neighborhood of $\widetilde{D}$. In particular, all local pseudo-holomorphic equations supplied by the regularized model remain unchanged.

It remains to check the coordinate statement. Let $p \in D$, and let $\widetilde{D}_{\alpha_1}, \dots, \widetilde{D}_{\alpha_m}$ be the branches of $\widetilde{D}$ through p, ordered so that the first q belong to D. In the regularized normal model at their common stratum there are J-holomorphic fiber coordinates $w_1, \dots, w_m$ with linearly independent differentials such that

$$\widetilde{D}_{\alpha_i} = \{w_i = 0\}, \qquad i = 1, \dots, m.$$

The smooth constant-rank theorem completes $(w_1, \dots, w_m)$ to smooth complex-valued coordinates $(z_1, \dots, z_n)$ centered at p, with $z_i = w_i$ for $i \le m$. After shrinking the coordinate neighborhood V,

$$D \cap V = \{z_1 \cdots z_q = 0\},$$

and $z_1, \dots, z_q$ are J-holomorphic.

The two deformations of the symplectic form in Steps I and III concatenate to a path of cohomologous symplectic forms along which D remains symplectic snc. Its initial form is the original symplectic form, while its terminal form is the form currently denoted by ω. Taking this terminal form as ω_1 gives the path in the Main Theorem. The rational form $\omega_{\mathbb{Q}}$ is auxiliary and does not belong to this path. □

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